

Statistics I

CSIT — 2nd Semester — 2079 — Answer Sheet
Subject Code: STA164

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. What do you understand by measures of central tendency and dispersion in descriptive statistics? What are the different measures of dispersion? The grouped frequency distribution in the table represents the masses of a sample of 50 of the people from the data file. "Brain size" Mass (lbs.) 1-22-33-44-55-66-77 Frequency 8 10 15 9 6 2 Compute mean, standard deviation, variance and coefficient of variation and interpret them.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Discuss the meaning of (i) positive (ii) negative (iii) perfect correlation between two variables. The following show the improvement (gain in reading speed) of 8 students in a speed reading program and the number of weeks they have been in program. Number of weeks 4 5 3 9 7 10 4 5 Speed gain (words per minute) 8 5 12 0 4 8 19 21 6 4 2 3 4 7 4 11 10 a. Compute correlation coefficient and interpret its result. b. Find the regression equation of speed gain on number of weeks. c. Estimate speed gain of a student who has been in program for 6 weeks and interpret the slope of the line.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. The joint density function of two continuous random variables X and Y is $f(x,y) = kxy$ $0 \leq x \leq 1, 0 \leq y \leq 1$ otherwise (a) Find the value of constant k. (b) Find $P(x > 3, y < 2)$ (c) Find $P(1$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions [8×5=40]

4. Differentiate between primary data and secondary data. What are the sources of secondary data?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What is sampling? Define simple random sampling and stratified random sampling with some relevant examples.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. What aspects of summary measures of data can be explained by the measures of skewness ? Kelvin Horn is the national sales manager for National Text Books. He has a sales staff of 10 who visit college professors all over the United States. Each Sunday morning he requires his sales staffs to send him a report. Listed below are the number of visits last week. 25610131521852030 Compute the five number summary.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. What are the requisites for good average? From the following distribution of marks 200 students of a college: Marks 30-40 40-50 50-60 60-70 70-80 80-90 No. of students 14 50 60 45 20 11 Compute (i) the minimum marks obtained by top 10% students. (ii) modal marks.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. The first four moments of a distribution about $x=2$ are 1, 2.5, 5.5, and 16. Calculate the first four moments about the mean. Test the skewness and kurtosis. Interpret the results.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Define independent and mutually exclusive events. Three groups of children contain respectively 2 boys and 2 girls, 3 boys and 1 girl, 1 boy and 3 girls respectively. One child is selected at random from each group. Find the probability of selecting one boy and two girls.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. What is conditional probability? Three roads A, B and C lead away from a jail. A prisoner escaping from the jail selects a road at random. If road A selected, the probability of escaping is $\frac{1}{10}$. Similarly for road B it is $\frac{1}{8}$ and for road C it is $\frac{1}{5}$. (i) What is the probability that the prisoner will succeed in escaping? (ii) If the prisoner had succeeded in escaping, what is the probability the he had chosen the road A?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Define binomial distribution. Under what conditions is the binomial distribution appropriate? The probability of a novice archer hitting the target with any shot is 0.3. Given that the archer shoots six arrows, find the probability that the target is hit at least one.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. State features of normal distribution. In a photographic process, the developing time of prints as a random variable having normal distribution with mean of 18.25 seconds with standard deviation 0.34 seconds. Find the probability that at least 17.64 seconds to develop one of the prints. Q.N.13 Define random variable. If X is the number of points rolled with a balanced die, find $E(X)$ and Variance of X. Also find expected value of random variable $g(Y) = 2X + 1$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.